Wearable fitness tracker to measure steps

instead of treatment and response.

In a recent development, IIIT-Delhi signed a Memorandum of Understanding (MoU) with the Lords Education and Health Society (Wish Foundation) with an aim to carry on research on health data analytics. For the coronavirus outbreak too, institutions such as World Health Organization (WHO) are utlilising data analytics and machine learning (ML) to track and predict its spread so that better decisions can be made.

According to a report from Research and Markets, global Big Data analytics market size in healthcare is expected to reach US$ 105.08 billion at a CAGR of 12.3 per cent during the forecast period of 2019 to 2027.

Data sources

In hospitals and other treatment centres, electronic health records (EHRs) are created in order to have separate comprehensive digital records for every patient containing such information as medical history, test reports, allergies and the like. These can be updated by a doctor even when the patient visits a different hospital and some new data needs to be added.

A large amount of this data consists of medical imaging reports, which can be analysed through algorithms that identify patterns in pixels based on previously available images to help with diagnosis. Last year, Google revealed its plans to provide search functionality to aid in performing tasks such as data entry and billing in these systems.

Apart from previous health data for statistics, real-time information related to vitals can be obtained from wearables and other health devices for population health management through predictive analytics. The IoT makes it possible to collect information using sensors in health equipment for remote monitoring. This is useful in identifying and prioritising people more at risk early on. The data integrated from various sources needs to be accurate for correct analysis.

What are we trying to achieve

Collecting and analysing large datasets and images through data analytics solution saves time by quickly filtering huge amounts of data to find accurate and customised solutions for tough problems that are beyond the scope of researchers and medical staff. Benefits are multifaceted and inter-related. Some of the major ones are discussed next.

Improved patient satisfaction and engagement. The aim of data analytics in any field is to make better decisions. As medical data gets analysed, better decision making and informed strategic planning by health professionals leads to improved patient satisfaction and recovery. Tedious and monotonous operations, such as making appointments, get streamlined. Researchers can analyse success rate of results from different treatments and filter out the ones that have better health outcomes. In the past, University of Florida made use of Google Maps and public health data to prepare neighbourhood-level hotspot maps for efficient delivery of primary care in most areas.

Healthcare dashboards can be used by healthcare professionals to monitor key performance indicators of patient statistics in real time. Health tracking devices like fitness bands make patients more aware about their health so that they take the necessary precautions and medications to prevent their problem from escalating. When a patient's parameters are alarming, the system immediately alerts the doctor to take necessary action(s) to deal with the situation.

Reduction in expenses. Big Data can be used to predict the admission rate of patients using ML. This is done by evaluating information including the number of patients present at a time, the costs incurred, total time of stay, among others to get an overall view of the process. This can help in understanding the right amount of hospital and clinic staff required so that the patients' care is proper and quick, with lower wait times and without unnecessary costs.

By leveraging population health data and identifying people that need more attention, patients can be kept away from hospitals. This reduces hospital expenses due to reduction in readmission rates.

Preventing fraud in insurance companies. Insurance companies provide high cover for small premium and are prone to fraud, such as in personal injury claims. Fraudsters often receive money into their accounts through identity theft. In other instances, they claim the records to be from leading hospitals and display more costly treatments than actually provided.

Using data to validate a prepayment and analytics to perform checks against public record databases ensures the validity of information provided to insurers. After verifying medical records, claims can be processed quickly so that genuine cases get resolved easily and treatment institutions are paid faster.